

3418. Maximum Amount of Money Robot Can Earn

Difficulty: Medium

Link: <https://leetcode.com/problems/maximum-amount-of-money-robot-can-earn/description/?envType=daily-question&envId=2026-04-02>

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Problem:

You are given an $m \times n$ grid. A robot starts at the top-left corner of the grid $(0, 0)$ and wants to reach the bottom-right corner $(m - 1, n - 1)$. The robot can move either right or down at any point in time.

The grid contains a value $coins[i][j]$ in each cell:

- If $coins[i][j] \geq 0$, the robot gains that many coins.
- If $coins[i][j] < 0$, the robot encounters a robber, and the robber steals the **absolute** value of $coins[i][j]$ coins.

The robot has a special ability to **neutralize robbers** in at most **2 cells** on its path, preventing them from stealing coins in those cells.

Note: The robot's total coins can be negative.

Return the **maximum** profit the robot can gain on the route.

Example 1:

Input: `coins = [[0,1,-1],[1,-2,3],[2,-3,4]]`

Output: 8

Explanation:

An optimal path for maximum coins is:

1. Start at $(0, 0)$ with 0 coins (total coins = 0).
2. Move to $(0, 1)$, gaining 1 coin (total coins = $0 + 1 = 1$).
3. Move to $(1, 1)$, where there's a robber stealing 2 coins. The robot uses one neutralization here, avoiding the robbery (total coins = 1).
4. Move to $(1, 2)$, gaining 3 coins (total coins = $1 + 3 = 4$).
5. Move to $(2, 2)$, gaining 4 coins (total coins = $4 + 4 = 8$).

Example 2:

Input: coins = $[[10,10,10],[10,10,10]]$

Output: 40

Explanation:

An optimal path for maximum coins is:

1. Start at $(0, 0)$ with 10 coins (total coins = 10).
2. Move to $(0, 1)$, gaining 10 coins (total coins = $10 + 10 = 20$).
3. Move to $(0, 2)$, gaining another 10 coins (total coins = $20 + 10 = 30$).
4. Move to $(1, 2)$, gaining the final 10 coins (total coins = $30 + 10 = 40$).

Constraints:

- `m == coins.length`

- `n == coins[i].length`
- `1 <= m, n <= 500`
- `-1000 <= coins[i][j] <= 1000`